Biological knowledge shortfalls of Laboulbeniomycetes as a case study for poorly studied taxa

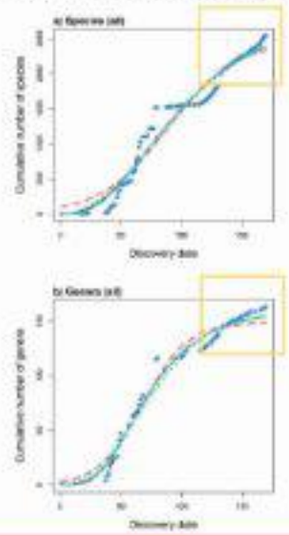
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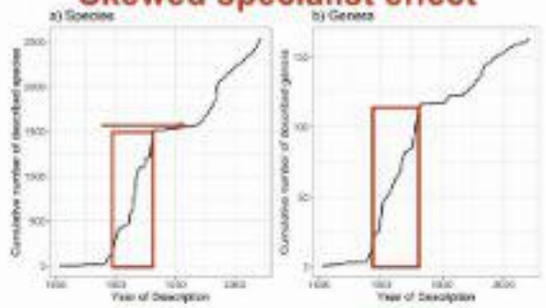
[1]Ghent University, Belgium; [2]University of Colorado Boulder, USA; [3]University of Birmingham, UK; [4]University of the Azores, Portugal; [5]Université du Québec en Abitibi-Témiscamingue, Canada; [6]Bangor University, UK

Laboulbeniomycetes (Labouls): fungi obligatorily associated with arthropods, historically neglected due to partical constraints, recent work based on DNA indicates high levels of cryptic diversity. How does being "understudied" affect knowledge shortfalls?

Linnean Shortfall (undescribed species)

In Labouls, increased sampling effort still returns new species, and the "actual" number of species is likely even higher than previously estimated



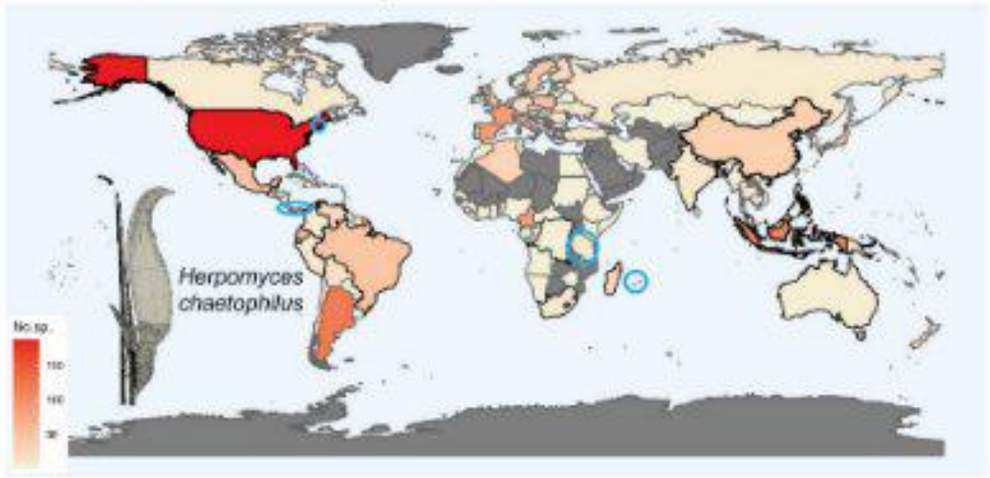






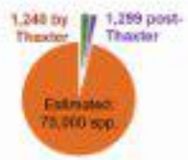
Wallacean Shortfall (distribution completeness)

We see a distributional unevenness in Labouls, and many taxa have disjunct distributions



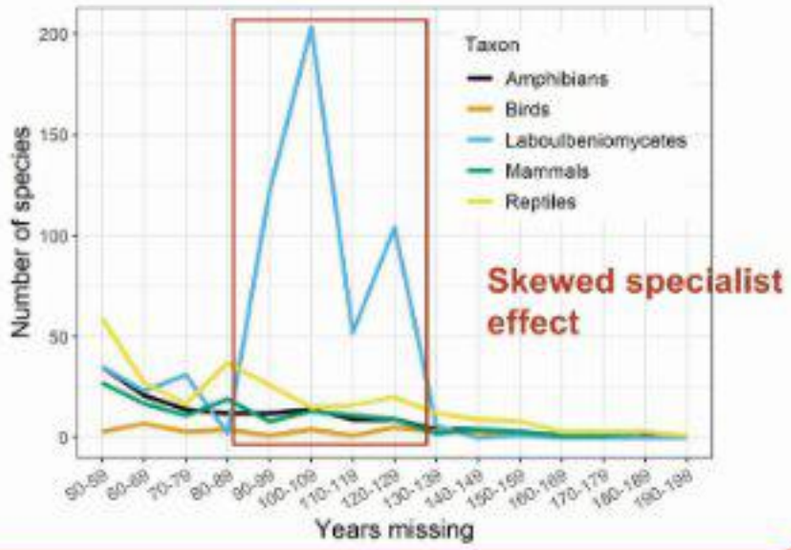
Latimerian Shortfall ("last seen")

Half of Labouls have no observations in 50+ years after description





Dr. Roland Thaxter
Active 1891–1931
92–132 years ago

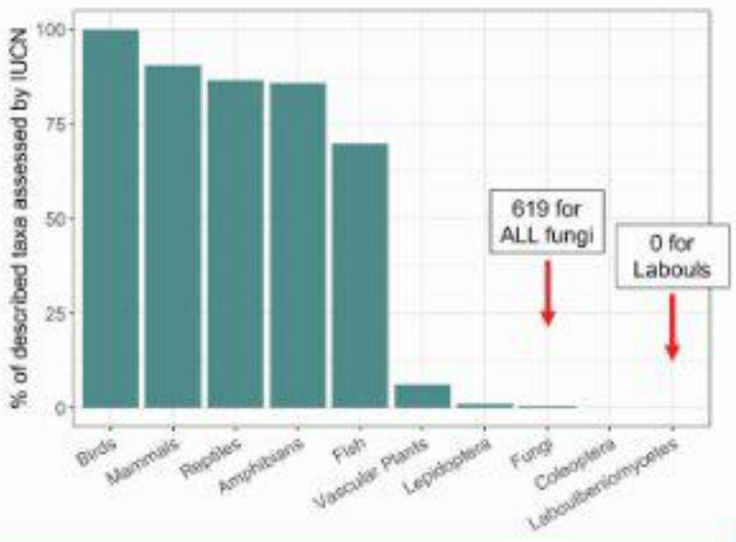


Scottian Shortfall (IUCN assessments)

No Red List assessments have been made for Labouls

Peter Markham Scott
(1909–1989), British conservationist





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